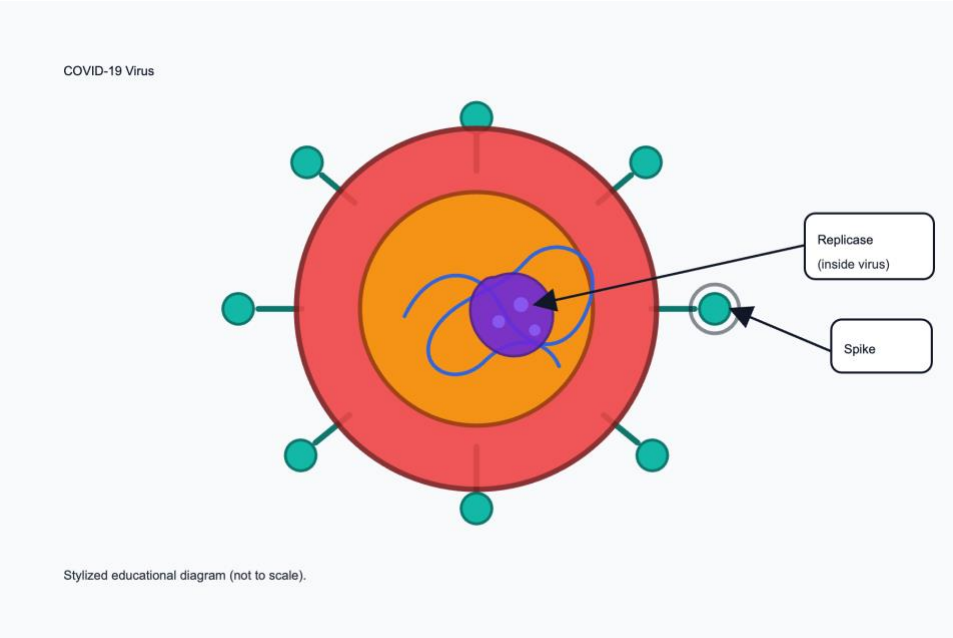
COSC 1437, Professor Clark
Lab 9 RSCU Analysis: 24 points (16 Part A, 8 Part B)

Write a Program: COVID-19 RSCU Analysis

Your goal is to perform Relative Synonymous Codon Usage (RSCU) analysis on specific regions of the COVID-19 virus genome (technical name*: severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2)*. In order to understand the terminology within these requirements, read the lab09-definitions.pdf file in the lab09-needed-files.zip (located within the Canvas Assignment where you found this document).

Program Description:

Your team will write a program for a genomic analysis tool to quantify Relative Synonymous Codon Usage (RSCU) within the SARS-CoV-2 genome. By comparing the replicase region (the virus's internal replication machinery) with the spike protein region(its external interface for host entry), you will provide mathematical reports regarding the codon bias both within and across the regions. This analysis allows you to determine if the virus maintains a consistent genetic signature or adapts its codon preferences between these two different proteins.



To complete this analysis, you will parse raw FASTA sequences from a COVID-19 genome sequenced in Texas in February 2026. I have already taken that entire genome and provided just the FASTA files for the replicase and spike regions. You will need to design an object-oriented solution to aggregate codon frequencies, calculate expected versus observed usage, and determine RSCU values for both genomic regions. Your final output will generate three detailed comparative datasets and then a formatted report that categorizes codons based on their favored status. This can give us insight into identifying significant shifts in usage between the internal replicase and the external spike protein.

Design Requirements and Suggestions:

You are provided with the following files to begin (located in Lab9NeededFiles.zip):

- codons.csv – a file of all codons with their sequence, amino acid name, and amino acid one-letter code
- replicase.fasta – the replicase region of the COVID-19 genome (bp 212-13420)
- spike.fasta – the spike protein region of the COVID-19 genome (bp 21548-25369)
- RSCU_formula.pdf – a detailed description of the RSCU calculation
- lab09-definitions.pdf – a list of vocabulary terms used within the project
- project-design.pdf – a (hopefully) helpful project plan for your team if you wish to use it

You're going to need a minimum of 2 classes but may choose to create as many as you wish. You'll need one for a CodonEntry and a Main class. If you go this design route, your Main class will be huge. This is fine, as long as there are many well documented methods in it. Or you may find the design to be cleaner with more classes. It's completely up to you.

Design this before you start trying to code it! With multiple team members, communicating who is completing which parts is crucial, and determining how those parts will interact with each other ahead of time will make the merging of your code pieces much smoother.

Minimally, you should consider holding the following information for your data type class design:

- CodonEntry
    - Codon sequence
    - Amino Acid name
    - Single letter AA abbreviation
    - A count of the codons in the replicase
    - A count of the codons in the spike
    - RSCU for the replicase
    - RSCU for the spike

You'll be handling a lot and will likely have many arrays/ArrayLists/variables/methods/etc. being managed in your Main class (or additional classes you choose to create). Be sure to document everything well (Javadoc comments, comments within code as needed).

Let me be clear, ***this project will not be easy***. It will take your whole group working together to figure out how you want to tackle this.

Output Requirements:

You will generate **three** separate csv files:

1. RSCU Analysis for Replicase Region (replicase_rscu.csv)
2. RSCU Analysis for Spike Region (spike_rscu.csv)
3. Comparative RSCU Analysis Report (rscu_comparison.csv)

And **one** formatted report:

1. Spike Favored Codons (spike_favored.txt)

For the individual replicase and spike analysis csv files, include for each codon:

- The codon code
- The full name of the amino acid
- The one letter code of the amino acid

- The total count of that codon within the fasta file
- The total count of all codons that make up the amino acid
- The percentage for which the specific codon occurs for the amino acid
- The RSCU value for the codon

For the comparative analysis csv, include:

- The codon code
- The full name of the amino acid
- The one letter code of the amino acid
- The RSCU value for the codon from the replicase file
- The RSCU value for the codon from the spike file
- The difference between the RSCU values. Use spike minus replicase RSCU values.
- The favored categorization for replicase
- The favored categorization for spike
- A column containing UP or DOWN if the category changes between the two
- NOTE: The category labels for the RSCU values are located within the RSCU_formula.pdf file

For the Spike Favored Codons report, first list the UP codons. You will list the codon, its amino acid, its replicase category, and its spike category. Then, provide the same information for the DOWN codons.

Submission Requirements:

Upload a .zip file for your group on Canvas containing the following items:

- A folder containing all of your code for the project
- RSCU Analysis Report for Replicase Region (replicase_rscu.csv)
- RSCU Analysis Report for Spike Region (spike_rscu.csv)
- Comparative RSCU Analysis Report (rscu_comparison.csv)
- Favored Spike Codon Report (spike_favored.txt)

Example Output:

I will not provide example output for the COVID-19 genome you are analyzing. However, I have provided a .zip file on Canvas (in our Chapter 9 module) with the requested generated reports on the H1N1 swine flu genome. You may look at those reports to see the format etc. of what you need to provide.

A note: you may wish to write your project first against the H1N1 genome. If you can get that to work and get my output, then it will only require very minor updates to your code to get it to work for COVID-19.

For the H1N1 FASTA files, pb1 is the replicase equivalent, and ha is the spike equivalent.